RESUME

Ignacio Cutignola

Buenos Aires, Argentina

Firmware & Embedded Systems Developer · Electronics Engineering Student (ITBA)

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PROFILE

Electronics Engineering student at ITBA (85% of the degree completed) and Electronics Technician. I currently develop firmware and hardware for embedded systems at Prosegur, with previous experience in cybersecurity (Accenture) and full-stack development (Balanz). As Director and Tutor at the IEEE ITBA Student Branch, I designed and taught programming courses that reached 430 students. I am looking to keep growing in firmware, embedded systems, and software development roles.

WORK EXPERIENCE

Semi-Senior Project Analyst | Prosegur

02/2026 - Present

- Firmware development for alarm panels, sensors, access control, and IoT devices on AVR (ATmega) and ESP32/ESP8266 microcontrollers.
- PCB design in Altium Designer and integration of UART, SPI, I2C, RS-485/Modbus, Wi-Fi, and Ethernet communications.
- Testing, validation, and commissioning of devices, providing technical support during installation.
- Supervision of a technical team of up to 3 people and management of development projects.

Tools: Altium Designer, Atmel Studio, Arduino, Python, BASCOM, C, C++, Fusion 360

Junior Cybersecurity Analyst | Accenture

07/2025 - 01/2026

- Cybersecurity incident response for telecommunications networks.
- Detection and analysis of alerts and events to determine their scope and impact.
- Incident documentation and preparation of technical reports.
- Work in Linux environments (Kali Linux and Ubuntu).

Tools: Kali Linux, Ubuntu

Junior Full-Stack Developer | Balanz

01/2022 - 07/2023

- Full-stack development of the investment platform, internal back-office systems, and API integrations using Node.js, Python 3, Angular, and TypeScript.
- Management of SQL, MongoDB, and Redis databases, improving query and service performance.
- Automation of internal processes and reports, and development of new features alongside business teams.
- Production support: diagnosing and fixing recurring incidents.
- Agile teamwork (Scrum) using Git and API testing with Postman.

Tools: Node.js, Python, Angular, TypeScript, SQL (PostgreSQL), MongoDB, Redis, Postman

Director and Tutor | IEEE ITBA Student Branch

03/2020 - 07/2023

- Designed and taught introductory and advanced courses on Python 3, TypeScript, and IoT, reaching 430 students during the pandemic.
- Created the syllabus, study materials, and exercises for each course.
- Coordinated a team of up to 10 volunteer tutors.
- Managed branch operations: enrollment, outreach, and events.

Recorded class: youtube.com/watch?v=AAAT5VprKgg

Tools: TypeScript, Python

Intern | Baxo S.A.

12/2019 - 12/2020

- Configuration of routers, switches, and wireless networks (Wi-Fi and point-to-point links).
- Work with fiber optics, structured cabling, and IP telephony (VoIP).
- Technical support for clients: diagnosing and resolving network failures.

EDUCATION

Electronics Engineering | Instituto Tecnológico de Buenos Aires — 2017 - Present

85% of the degree completed

Electronics Technician | Escuela Cristiana Evangélica Argentina — 2011 - 2016

TECHNICAL SKILLS

Embedded & Hardware: C, C++, Assembly, Atmel Studio, BASCOM, Arduino, Raspberry Pi, Altium Designer, LTspice, EPLAN

Software & Data: Python, TypeScript, Node.js, Angular, SQL (PostgreSQL), Redis, MongoDB, Postman

Systems & Security: Linux, Ubuntu, Kali Linux

Design, Computing & Documentation: Fusion 360, Blender, MATLAB, LaTeX, Photoshop, Premiere Pro, Excel

LANGUAGES

Spanish: Native **English:** Upper-Intermediate (B2) **Portuguese:** Intermediate

HONORS & AWARDS

ITBA Honors Scholarship

PROJECTS

Color classifier — Academic project

Team-built machine that sorts products by color onto a rotating disc, with automatic and manual modes. It integrates a DC-motor conveyor belt, a color sensor, an inductive sensor for homing, infrared sensors for detection and arrival, a stepper motor, Arduino Nano control, an operator safety panel, electrical panel drawings designed in EPLAN, and 3D-designed and modeled parts.

LinkedIn post: lnkd.in/p/dekPENSN

Arduino · C · Altium Designer · EPLAN · Fusion 360

Signal sampling and recovery simulator — Academic project

Python desktop application (PyQt5, NumPy, SciPy, and Matplotlib) that simulates the full sampling and reconstruction chain of an analog signal. It models an 8th-order anti-aliasing filter (Legendre approximation) built from 4 Sallen-Key cells defined by real components, a sample & hold, an analog switch with configurable duty cycle, and a recovery filter. Each stage can be enabled or bypassed and is displayed in time and frequency (FFT), on an extensible architecture with a common base class.

GitHub repository: github.com/cutignolai/ASSD-TP1

Python

Robotic arm that draws from images — Academic project

Together with my university team, we implemented a robotic arm that receives an image of a drawing and is able to interpret and replicate it.

MATLAB · Computer Vision

Pulp Vision 3D — Personal venture

3D printing venture: design and manufacturing of mechanical parts and custom spare parts on an Ender 3 SE, in PLA and PETG, from modeling to sale.

Fusion 360 · Blender · Photoshop · Premiere Pro · Excel

Home stock manager — Personal project

Home system that registers products with a barcode reader and uses a Raspberry Pi as its central brain: it tracks stock, sends email alerts when something runs out, and builds the shopping list.

Arduino · Raspberry Pi · MongoDB · SQL (PostgreSQL) · Linux

Private lessons

Private tutoring

Lessons in Physics, Mathematics, C/C++, and Python 3 for a group of 5 to 10 students, with a focus on conceptual clarity and personalized support.

Python C++